RESTRICTION OF THE NORMALIZATION COVER TO A DENSE OPEN WHERE THE MORPHISM IS FINITE

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Normalization factorization. lemma-there-is-a-scheme-integral-over (carded L26715). The integral cover with local factorization through U.
- Finiteness over an open. Morphisms, lemma quasi-finite-finite-over-open (cross-chapter, named). Quasi-finite over a quasi-compact open is finite after shrinking.
- Schematic image of open immersions. Morphisms of Spaces, lemma composition-integral (cross-chapter, named). Integrality of the assembled morphism.

Source reference. Original lemma 0ABV: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 4792–4792. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0BB9: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 4911–4911. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0ACZ: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 4979–4979. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0AD0: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 5053–5053. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BAI: focused-seeds/geometry/S9-decent-finite-over-dense-open/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 02NW: focused-seeds/geometry/S9-decent-finite-over-dense-open/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0AAY: focused-seeds/geometry/S9-decent-finite-over-dense-open/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NZ: focused-seeds/geometry/S9-decent-finite-over-dense-open/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NX: focused-seeds/geometry/S9-decent-finite-over-dense-open/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BAQ: focused-seeds/geometry/S9-decent-finite-over-dense-open/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BAR: focused-seeds/geometry/S9-decent-finite-over-dense-open/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03FZ: focused-seeds/geometry/S9-decent-finite-over-dense-open/ref-tags.txt, source lines 12–12. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f is locally of finite type. Let $X^0 \subset |X|$, resp. $Y^0 \subset |Y|$ denote the set of codimension 0 points of X, resp. Y. Assume*

(1) *Y is decent,*
(2) *X^0 and Y^0 are finite and $f^{-1}(Y^0) = X^0$,*
(3) *either f is quasi-compact or f is separated.*

Then there exists a dense open $V \subset Y$ such that $f^{-1}(V) \to V$ is finite.

Proof. By Lemmas 0BB9 and 0ABV we may assume Y is a scheme with finitely many irreducible components. Shrinking further we may assume Y is an irreducible affine scheme with generic point y. Then the fibre of f over y is finite.

Assume f is quasi-compact and Y affine irreducible. Then X is quasi-compact and we may choose an affine scheme U and a surjective étale morphism $U \to X$. Then $U \to Y$ is of finite type and the fibre of $U \to Y$ over y is the set U^0 of generic points of irreducible components of U (Properties of Spaces, Lemma 0BAQ). Hence U^0 is finite (Morphisms, Lemma 0AAY) and after shrinking Y we may assume that $U \to Y$ is finite (Morphisms, Lemma 02NW). Next, consider $R = U \times_X U$. Since the projection $s : R \to U$ is étale we see that $R^0 = s^{-1}(U^0)$ lies over y. Since $R \to U \times_Y U$ is a monomorphism, we conclude that R^0 is finite as $U \times_Y U \to Y$ is finite. And R is separated (Properties of Spaces, Lemma 03FZ). Thus we may shrink Y once more to reach the situation where R is finite over Y (Morphisms, Lemma 0BAI). In this case it follows that $X = U/R$ is finite over Y by exactly the same arguments as given in the proof of Lemma 0ACZ (or we can simply apply that lemma because it follows immediately that X is quasi-separated as well).

Assume f is separated and Y affine irreducible. Choose $V \subset Y$ and $U \subset X$ as in Lemma 0AD0. Since $f|_U : U \to V$ is finite, we see that $U \subset f^{-1}(V)$ is closed as well as open (Morphisms of Spaces, Lemmas 04NX and 04NZ). Thus $f^{-1}(V) = U \amalg W$ for some open subspace W of X. However, since U contains all the codimension 0 points of X we conclude that $W = \emptyset$ (Properties of Spaces, Lemma 0BAR) as desired. □